From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

FIITJEE

JEE(Main), 2014

ANSWERS, HINTS & SOLUTIONS

FULL TEST – IV
(Main)

S. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**C**	**A**	**A**
2.	**D**	**C**	**B**
3.	**C**	**B**	**A**
4.	**A**	**C**	**A**
5.	**D**	**D**	**B**
6.	**A**	**D**	**C**
7.	**A**	**D**	**C**
8.	**A**	**D**	**B**
9.	**C**	**D**	**B**
10.	**B**	**D**	**C**
11.	**A**	**A**	**C**
12.	**C**	**A**	**B**
13.	**C**	**C**	**B**
14.	**D**	**C**	**A**
15.	**C**	**B**	**B**
16.	**C**	**A**	**D**
17.	**D**	**D**	**A**
18.	**C**	**C**	**B**
19.	**A**	**D**	**A**
20.	**C**	**C**	**D**
21.	**C**	**D**	**D**
22.	**B**	**D**	**A**
23.	**C**	**C**	**C**
24.	**D**	**D**	**B**
25.	**D**	**A**	**A**
26.	**C**	**C**	**D**
27.	**B**	**D**	**A**
28.	**C**	**C**	**D**
29.	**C**	**A**	**D**
30.	**C**	**D**	**C**

Physics

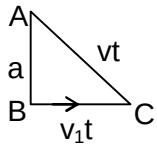
PART – I

SECTION – A

1. Zero after decimal point is also significant.

2. $a^2 + v_1^2t^2 = v^2t^2$

$$t = \frac{a}{\sqrt{v^2 - v_1^2}}$$



3. $600 - T = ma$

$$a_{min} = \frac{600 - 360}{60} = 4 \text{ m/s}^2$$

4. $$-\mu mgt_0 = \frac{mv_0}{2} - mv_0$$

$$\mu = \frac{v_0}{2gt_0}$$

5. Friction force on upper block is $f = ma$
So work done = $ma \times s$

7. $mu = m(v/2) + mv$

$\Rightarrow$ $v = (2/3)u$

$$-e = \frac{\frac{u}{3} - \frac{2u}{3}}{u}$$

$\Rightarrow$ $e = 1/3$

8. $$T = \frac{2\pi R_2}{\sqrt{\frac{GM}{R_2}}}$$

$$\Rightarrow \quad \frac{GM}{R_2^3} = \frac{4\pi^2}{T^2}$$

$$g = \frac{GM}{R_1^2} = \frac{4\pi^2R_2^3}{T^2R_1^2}$$

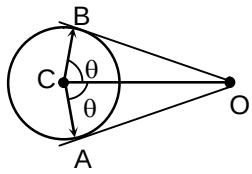
9. $$m\left(-\frac{dv}{dt}\right) = 6\pi\eta rv$$

$$-\int_2^{0.5} \frac{dv}{v} = 6\pi\left(\frac{1}{18\pi}\right)\int_0^t dt$$

$$\ell n\frac{2}{0.5} = \frac{t}{3} \Rightarrow t = 3\ell n4 \text{ sec}$$

12. $T = 2\pi\sqrt{\dfrac{\ell}{g_{eff}}}$ **where** $g_{eff} = \dfrac{mg - qE}{m} = g - \dfrac{qE}{m}$

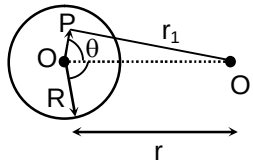
13. Maximum will be corresponding to source position A and minimum to source friction B.



So, time interval = $\dfrac{2\theta}{\omega} = \dfrac{2\cos^{-1}\left(\dfrac{3}{5}\right)}{5}$

18. $V_P = V_{pindued} + V_{Pq} = V_0$

$V_{pinduced} = \dfrac{kq}{r} - \dfrac{kq}{r_1}$



20. $v_i = \dfrac{1}{2}\dfrac{6}{5}CV^2$ …(i)

$q_i = \dfrac{6}{5}CV$ ….(ii)

$q_f = \dfrac{11}{5}CV$ …(iii)

$U_f = \left(\dfrac{1}{2}\dfrac{6}{5}CV^2 + \dfrac{1}{2}CV^2\right)$

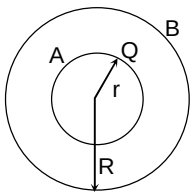
Charge flown from battery = CV

Work done = CV^2

Heat produced $\Delta H = \Delta U + \Delta W$

$= \left[\left(\dfrac{1}{2}\dfrac{6}{5}CV^2 + \dfrac{1}{2}CV^2\right) - \dfrac{1}{2}\dfrac{6}{5}CV^2\right] - CV^2 = -\dfrac{1}{2}CV^2$

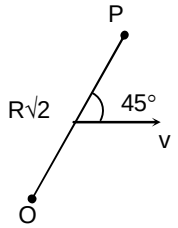
21. As A and B will be at same potential so charge flown equal to Q.



22. $B = \sqrt{B_A^2 + B_B^2} = \dfrac{\sqrt{2}\mu_0 I_0}{2\pi a\sqrt{2}} = \dfrac{\mu_0 I_0}{2\pi a}$

23. Using $e = \int \vec{B}\cdot\left(d\vec{\ell}\times\vec{v}\right)$

So, e = BvR



24. $e = \int_{-\ell/2}^{+\ell/2} B(x\omega)dx = 0$

25. $\frac{1}{2}\frac{q^2}{(C_1+C_2)} = \frac{1}{2}LI_0^2$

$q = \sqrt{L(C_1+C_2)I_0}$

$q_1 = \frac{q}{C_1+C_2}C_1 = C_1I_0\sqrt{\frac{L}{(C_1+C_2)}}$

27. $\frac{I_{coherent}}{I_{incoherent}} = \frac{4I}{2I} = 2$

29. Using Einstein equation

$E = W + eV$

$\frac{12400\ ev\,\overset{o}{A}}{0.2\times10^4\ \overset{o}{A}}$ = 4.6 eV + eV

V = 1.6 Volts

Chemistry

PART – II

SECTION – A

1. For H, $\Delta r_H = r_3 - r_2 = \frac{0.529}{1} \times [9-4]$

For Li^{+2}, $\Delta r_{Li^{+2}} = r_3 - r_2 = \frac{0.529}{3}[9-4]$

$\Rightarrow \frac{\Delta r_H}{\Delta r_{Li^{+2}}} = \frac{3}{1}$

2. $2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2$

$Cu_2O + FeS \longrightarrow FeO + Cu_2S$

$2FeS + 3O_2 \longrightarrow 2FeO + 2SO_2$

$FeO + SiO_2 \longrightarrow FeSiO_3\,(Slag)$

4. This is a 3° and resonance stabilized carbocation.

5. Cumulated diene is least stable, evolve max heat in the hydrogenation reaction.

11. Number of revolution made by an electron per sec

$$= \frac{v_n}{2\pi r_n} \approx \left(\frac{2\pi Ze^2}{nh \times 2\pi \left(\frac{n^2h^2}{4\pi mZe^2} \right)} \right)$$

$$= \frac{4\pi^2 mZ^2e^4}{n^3h^3} \approx \frac{4\pi^2 me^4}{h^3}\left(\frac{Z^2}{n^3}\right)$$

$$= \frac{Z^2}{n^3} \times 6.66 \times 10^{15} \approx \frac{1}{8} \times 6.66 \times 10^{15}$$

$= 8.3 \times 10^{14}$

13. Vapour pressure depends on temperature not on size of container.

14. There is no intermolecular forces of attraction in ideal gases.

16. $3A + B \longrightarrow 2C$

$$r = \frac{1}{3}\frac{d[A]}{dt} = -\frac{d[B]}{dt} = \frac{1}{2}\frac{d[C]}{dt} = K'[A][B]$$

$$-\frac{d[A]}{dt} = 3K'[A][B] = K[A][B]$$

i.e. $K = 3K'$

So that $\frac{d[C]}{dt} = 2K'[A][B] = \frac{2}{3}K[A][B]$

17. $K_P = P_{CO_2}$, which will depends temperature of the system only.

18. Water will be neutral as $[H^+] = [OH^-]$

20. Zinc is more electropositive than silver therefore, it displaces less electropositive silver.

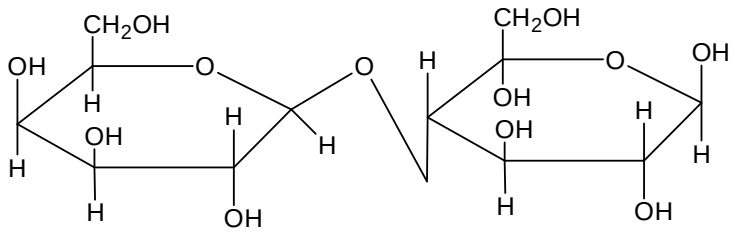
21. Carnalite is a double salt $KCl.MgCl_2.6H_2O$ which give test of K^+, Mg^{+2} and Cl^-.

22. $2Cu^{+2} + 2CN^- \longrightarrow 2Cu^+ + (CN)_2$

$Cu^+ + CN^- \longrightarrow CuCN \xrightarrow{3KCN} K_3[Cu(CN)_4]$

23. Mass number change only by α-particles and atomic number reduces by 2 unit by each α-particle and increase by 1 unit by β-particle.

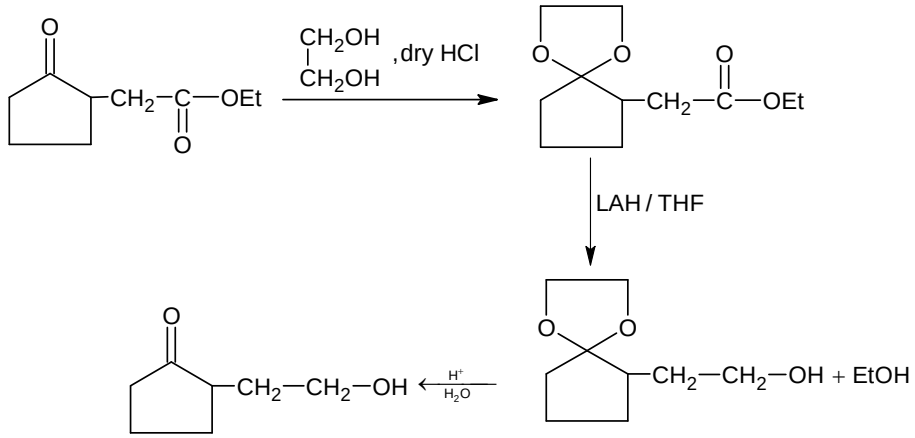
27.



Galactose unit Glucose unit

28. The 1st step is electrophilic addition i.e. rate determining step, more stable will be the intermediate better will be the rate of reactions.

29.



Mathematics

PART – III

SECTION – A

1. $(HHH),(RR),\underbrace{(I,I),(P,P),AYU}_{x} = {}^{12}C_7 \cdot \frac{7!}{2!2!} \cdot 1 = (198)7!$

2. $|(a+b)(a-b)| = |a-b|^2 \Rightarrow |a+b| = |a-b|$

$\Rightarrow \left|\frac{a}{b}+1\right| = \left|\frac{a}{b}-1\right|$

$\Rightarrow \frac{a}{b}$ is perpendicular bisector of (–1, 0) and (1, 0)

$\therefore \frac{a}{b}$ is purely imaginary

3. $\frac{dV(t)}{dt} = -K(T-t)$

$\Rightarrow V(t) = \frac{K(T-t)^2}{2} + C$

At t = 0, V = I $\Rightarrow C = I - \frac{KT^2}{2}$

$\therefore$ Scrap value V(T) $\Rightarrow$ V(t = T) = C = $I - \frac{KT^2}{2}$

4. $2(\tan^{-1}x)^2 - \pi(\tan^{-1}x) - \frac{3\pi^2}{8} = 0$

$\Rightarrow \tan^{-1}(x) = -\frac{\pi}{4}$

$\therefore$ x = –1

5. $\lim\limits_{t\to\infty} \frac{(t^2-2)}{t^2\left(\sqrt{2-\frac{1}{t}-\frac{1}{t^2}}+\sqrt{1-\frac{1}{t}+\frac{1}{t^2}}\right)(\sqrt{2}-1)} = 1$

Now ||x – 1| – 6| = k has four distinct solutions if k $\in$ (0, 6)

$\therefore$ Number of integral values of k is 5

6. $f(x) = \frac{x|x|}{2}$

7. **Case–I:** when $\frac{p-2}{2} \le 2 \Rightarrow p \le 6$ and f(2) = 8

$\therefore$ P = 2

Case–II: when $\frac{p-2}{2} > 2 \Rightarrow p > 6$

$-\frac{D}{4a} = 8 \Rightarrow -\frac{(p-2)^2 - 4(3p-2)}{4} = 8$

$\Rightarrow P = 8 + 2\sqrt{5}$

$\therefore\ p = 8 + 2\sqrt{5}$ and $p = 8 - 2\sqrt{5}$ (rejected)

$\therefore$ Sum of values of p is $2 + 8 + 2\sqrt{5} = 10 + 2\sqrt{5} = 10 + \sqrt{20}$

$\therefore$ m + n = 30

8. We have $(x - y)^2 + (y - 3)^2 = 0$

$\therefore x = y = 3$

9. We have $f(x) > f(2)\ \forall\ x < 2$

$\Rightarrow \lim\limits_{x \to 2^-} f(x) \geq f(2) = 1$

$\Rightarrow a^2 - 9a - 9 \geq 1$

$\Rightarrow a^2 - 9a - 10 \geq 0$

$\Rightarrow a \leq -1$ or $a \geq 10$

But a is positive. Hence $a \geq 10$. That is $a \in [10, \infty[$

Note that when a = 10

$$f(x) = \begin{cases} 3 - x, & x < 2 \\ 2x - 3, & x \geq 2 \end{cases}$$

So that f is continuous at 2, $f'(2)$ does not exist and $f'(x) \neq 0\ \forall\ x \in R$. Therefore, x = 2 is a critical point and $f'(x)$ changes sign from negative to positive at x = 2

10. We have $a = b + \frac{1}{a + \frac{1}{a}}$

$\Rightarrow (a - b) = \frac{1}{a + \frac{1}{a}}$ and $b = a - \frac{1}{b + \frac{1}{b}}$

$\Rightarrow a - b = \frac{1}{b + \frac{1}{b}}$

$\therefore\ a + \frac{1}{a} = b + \frac{1}{b} \Rightarrow (ab - 1)(a - b) = 0$

$\therefore ab = 1\ \because\ a \neq b$

$\therefore\ (a - b) = \frac{1}{a + b} \Rightarrow a^2 - b^2 = 1$

11. Probability that calculator of brand r is selected and is defective

$= \sum_1^6 (kr)\left(\frac{7 - r}{21}\right) = \frac{k}{21}\sum_{r=1}^{6}(7r - r^2) = \frac{8k}{3}$ ..... (1)

$\Rightarrow$ Let E_r denote the event that calculator of brand r is selected $P(E_r) = kr$

Since E_r(r = 1, 2, ..., 6) are mutually exclusive and exhaustive events we must have

$\sum_{r=1}^{6} P(E_r) = 1 \Rightarrow \sum_{r=1}^{6} kr = 1 \Rightarrow k = \frac{1}{21}$

$\therefore$ required probability = $\frac{8k}{3} = \frac{8}{63} = \frac{p}{q}$

$\therefore$ (p + q) = 71

12. $P(r) = (2r + 1)^2$

$$\sum_{r=1}^{n-1}\sqrt{P(r)} = \sum_{r=1}^{n-1}(2r+1) = n^2 - 1$$

$$\therefore \frac{1}{2} = \lim_{n\to\infty}\frac{n^2-1}{an^3+bn^2+c}$$

$\Rightarrow a = 0, b = 2$

So, $(a + b) = 2$

13. Let $\tan^{-1} x = \alpha$ and $\tan^{-1} y = \beta$ then $a \tan\alpha + b \sec\alpha = c$ and $a \tan\beta + b \sec\beta = c$

Obviously $a \tan\theta + b \sec\theta = c$ has roots $\tan\alpha$ and $\tan\beta$

$\Rightarrow (a^2 - b^2) \tan^2\theta - 2ac \tan\theta + c^2 - b^2 = 0$

$$\therefore \tan\alpha + \tan\beta = \frac{2ac}{a^2-b^2} \text{ and } \tan\alpha\tan\beta = \frac{c^2-b^2}{a^2-b^2}$$

$$\text{So } \frac{x+y}{1-xy} = \frac{2ac}{a^2-b^2}$$

14. $$\lim_{x\to o^+} e^2 \frac{\left(5^{\left(\frac{x(x-2)}{2}\right)}-1\right)}{\frac{2x(x-2)}{2}\frac{\tan x}{x}} = \frac{1}{2}e^2\ln 5$$

$$\lim_{x\to o^-}\frac{\lambda\left(e^2\left((\sqrt{5})^x-1\right)\right)\left(e^{x^2}-1\right)}{x\frac{2\sin x}{x}\left(\frac{1-\cos x}{x^2}\right)x^2} = \lambda e^2\ln\sqrt{5} = \lambda\frac{e^2}{2}\ln 5$$

$\therefore \lambda = 1$

15. $$\left(\frac{1-3\tan^2 1^o}{3-\tan^2 1^o}\right)\frac{\cot 1^o}{\cot 3^o} = \left(\frac{1-3\tan^2 1^o}{3\tan 1^o - \tan^3 1^o}\right)\frac{1}{\cot 3^o} = \frac{1}{\tan 3^o \cot 3^o} = 1$$

16. $$\lim_{x\to 0}\frac{(1+\sin 2x)^{1/x}}{1} = e^2$$

17. $f(x) = x^2 + 3x - 2$

$$\therefore \int f(x)\,dx = \frac{x^3}{3} + \frac{3x^2}{2} - 2x + c$$

18. $f(x) = (x - 1)^4 (x - 2)^3 (x - 3)^2 (x - 4)$

$f'''(1) = f''(2) = f'(3) = 0$

$$f'(x) = (x-1)^4(x-2)^3(x-3)^2\cdot 1 + (x-1)^4(x-2)^3\cdot 2(x-3)(x-4)$$
$$+(x-1)^4\,3(x-2)^2(x-3)^2(x-4) + 4(x-1)^3(x-2)^3(x-3)^2(x-4)$$

$\therefore f'(4) = 3(6^3)$

19. $a + ar = 12$ ..... (1)

$ar^2 + ar^3 = 48$ ..... (2)

$\therefore r^2 = 4 \Rightarrow r = -2, \because r \neq -1, 2$

20. $P\left(\frac{3k+2}{k+1}, \frac{5k+3}{k+1}, \frac{6k+4}{k+1}\right)$

$\therefore$ Hence $\frac{3k+2}{k+1} = \frac{13}{5} \Rightarrow k = \frac{3}{2}$

21. $\Delta = \begin{vmatrix} 1 & 2a & a^2 \\ 1 & 2b & b^2 \\ 1 & 2c & c^2 \end{vmatrix} \cdot \begin{vmatrix} 1 & p & p^2 \\ 1 & q & q^2 \\ 1 & r & r^2 \end{vmatrix} = 2 \cdot 2\Delta_1 \cdot 2\Delta_2 = 8\Delta_1\Delta_2 = 8 \cdot \frac{1}{2} \cdot 4 = 16$

22. $(a + b\lambda)x + (2b - 2a\lambda)y + (3b - 3\lambda a) = 0$

$\because\ a + b\lambda = 0 \Rightarrow \lambda = -\frac{a}{b}$

$\therefore\ y = -\frac{3}{2}$

23. $\frac{\left(\frac{x-y+1}{\sqrt{2}}\right)^2}{10} + \frac{\left(\frac{x+y-3}{\sqrt{2}}\right)^2}{5/2} = 1$

Here $a^2 = 10$ and $b^2 = 5/2$ and centre is (1, 2)

$\because$ Locus of feet of perpendicular lie on auxiliary circle of ellipse

$\therefore$ Equation of circle is $(x - 1)^2 + (y - 2)^2 = 10$

24. $f(x) = \begin{cases} 4; & 0 < x < \frac{\pi}{2} \\ -2 & \frac{\pi}{2} < x < \pi \\ 0; & \pi < x < \frac{3n}{2} \\ -2 & \frac{3\pi}{2} < x < 2\pi \end{cases}$

$\therefore$ Range of function = {0, –2, 4}

25. $\sum \tan^2\left(\frac{\theta}{2}\right) = \sum \frac{1-\cos\theta}{1+\cos\theta} = \sum \left(\frac{1-\frac{a}{b+c}}{1+\frac{a}{b+c}}\right)$

$= \sum \left(\frac{b+c-a}{a+b+c}\right) = \frac{a+b+c}{a+b+c} = 1$

26. $3^4 = 81$

27. We have, $\cos^2 x - (c - 1)\cos x + 2c - 6 \geq 0\ \forall\ x \in R$

$\Rightarrow (\cos x - 2)(\cos x - (c - 3)) \geq 0$

$\therefore \cos x \leq c - 3$

$\Rightarrow c - 3 \geq 1 \Rightarrow c \geq 4$

28. $px^2 + qx + r = 0$
$rx^2 + qx + p = 0$ (on subtract)
$\therefore x = -1$
So common root is – 1
$\therefore p - q + r = 0$

29. $[\{x\}] = 0$
$f(x) = (3 - x^7)^{\frac{1}{7}}$
$f(f(x)) = \left(\left(3 - (3 - x^7)^{\frac{1}{7}}\right)^7\right)^{\frac{1}{7}} = x$
$f(f(x)) = x \Rightarrow f^{-1}(x) = f(x)$
$\therefore f^{-1}(50) = f(50)$ and $f(f(100)) = 100$
$\therefore f^{-1}(50) - f(50) + f(f(100)) = 100$

30. Put $a = 2R \sin A$, $b = 2R \sin B$
$2R \sin B (\sin^2 A + \cos^2 A) = \sqrt{2}a$
$\therefore 2R \sin B = \sqrt{2}a$
$\Rightarrow \left(\frac{b}{a}\right) = \sqrt{2}$